IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 17 April 2024, accepted 8 June 2024, date of publication 12 June 2024, date of current version 19 June 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3412948

RESEARCH ARTICLE

Mountain Flood Level Forecasting in Small Watersheds Based on Recurrent Neural Networks and Multi-Dimensional Data

SONGSONG WANG[1,2] **AND OUGUAN XU**[1]

[1]College of Information Engineering, Zhejiang University of Water Resources and Electric Power, Hangzhou 310018, China
[2]College of Engineering, Ocean University of China, Qingdao 266100, China

Corresponding author: Songsong Wang (wss@zjweu.edu.cn)

This work was supported in part by the Humanities and Social Science Fund of Ministry of Education of China, "Research on the Probability Deep Learning Early Warning Decision Scheme for Mountain Floods Considering Social Factors" under Grant 22YJCZH162; in part by the Joint Funds of Zhejiang Provincial Natural Science Foundation of China under Grant LZJWZ23E090001; in part by Zhejiang Provincial Education Science Planning 2023 General Planning Project (University), "Research on the Ideological and Political System of Artificial Intelligence Courses in Universities with Industry Characteristics under the Integration of Professions and Industries" under Grant 2023SCG382; in part by the Key Laboratory for Technology in Rural Water Management of Zhejiang Province by Zhejiang University of Water Resources and Electric Power, China, under Grant ZJWEU-RWM202107; and in part by the Research Center for Digital Economy and Sustainable Development of Water Resources by Zhejiang University of Water Resources and Electric Power, China, under Grant xrj2022013.

ABSTRACT Due to the many disaster-causing factors of mountain torrents, rapid changes in time and space, and rapid disasters, the traditional hydrological models and general data-driven mountainous flood forecasting models in small watershed areas have low accuracy and timeliness. Therefore, this paper proposed the time series Recurrent Neural Networks (RNN) and data processing for small mountain flood level forecasting, it includes base RNN, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) model. We use the Tiaoxi catchment's mountain flood data including hydrology, meteorology, geography and human activities at Zhejiang province of China. We construct multi-dimensional time series data processing and the model test system, and develop multi-dimensional disaster factor data on an hourly scale to suit the training and prediction of Deep Learning (DL)'s structure. The experiment shows that time series base RNN, LSTM and GRU models can be considered as the efficient hydrology modeling approach. The LSTM model demonstrated the best accuracy and real-time in the study area.

INDEX TERMS Mountain flood forecasting, small watersheds, RNN, LSTM, multi-dimensional data, deep learning.

I. INTRODUCTION

As rainfall, topography, and human activities are the main factors leading to mountain torrent disasters, the mechanisms behind the formation and causes of heavy rain and floods in mountainous areas are still incomplete, which result in challenges in the development of hydrological and hydrodynamic forecasting and early warning technology systems [1], [2]. In addition, the flood forecasting method is mainly applied to large river basins, it does not meet the requirements in terms of timeliness for forecasting sudden and strong flash floods in small watersheds. The rainfall observations inherently have certain errors and probabilities, and the predicted mountain flood disasters also have certain uncertainties. In recent years, the flood forecasting is adopted mainly by hydrological and hydrodynamic models [3], [4]. Hydrological and hydrodynamic models are difficult to make correct judgments on decision-making risks [5], [6]. Moreover, in the field of small watershed research, the ability to obtain information and forecast mountain flood level is limited, and the forecasting effect is not obvious. The advantages of data-driven artificial intelligence (AI) forecasting have not been fully explored.

Artificial Neural Network (ANN) [7] is especially known to well mimic highly non-linear and complex systems, many

The associate editor coordinating the review of this manuscript and approving it for publication was Khursheed Aurangzeb.

© 2024 The Authors. This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

studies applied ANNs for modelling runoff processes [8]. ANNs are a group of classical data-driven methods that have wide applications in many regions and have gained considerable attention, due to their capability to uncover nonlinear relationships in time series forecasting with reasonably reliable accuracy. For example, ANNs have been used in daily runoff forecasting [9], water level forecasting [10], water quality simulation [11], monthly flow forecasting [12], and precipitation forecasting [13]. Among these, Deep Learning (DL)'s Recurrent Neural Networks (RNN) and its improved version have captured much attention in the field of time series forecasting, because the structural design of its hidden state is parallel to the time series data characteristics or any sequence data [14]. Furthermore, it also advances the research in hydrology [15], [16].

This study aims to develop a real-time and accurate regional mountain flood forecasting model based on the DL and data analysis techniques, such that actions can be efficiently improve the accuracy and real-time performance of forecasting mountain flood water levels in small watersheds. This study proposed the time series DL model to mountain flood level forecasting according to Recurrent Neural Networks (RNN), it includes base RNN, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) model. We use the Tiaoxi catchment's mountain flood data including hydrology, meteorology, geography and human activities at Zhejiang province of China. We construct multi-source heterogeneous data processing and the model test system, and develop multi-dimensional disaster factor data on an hourly scale to suit the training and prediction of DL model. The experiment shows that time series RNN, LSTM and GRU models can be considered as the efficient hydrology modeling approach. The LSTM model demonstrated the best accuracy and real-time in the study areas, and we underscored the potential of the proposed DL models for mountain flood level forecasting in small watersheds.

This is how the remainder of the paper is structured. Section III presents the related works. Section III presents our forecasting mode using big data and DL. The followed by Section IV which give the study area of complex small watersheds at the hourly time scale and data collection and analysis. Section V presents our model results and the benchmark results, and discusses the implications of the results. Section VI presents the discussions about data, models and generalizability, and compare with other literature achievements to obtain the advantages and disadvantages of the proposed method in this article. The conclusions of this study are presented in Section VII.

II. RELATED WORKS

Flood level forecasting method machine learning [17], [18], DL [19], integrated model [20], [21] and the Complex models [22]. The LSTM neural network method is an improved version of a typical RNN, which equipped with the capacity to retain long time dependencies [23]. In recent years, this method is increasingly topical in water conservancy. Zhang et al. [24] used the LSTM to predict the daily water level in agricultural areas. In another study, Hu et al. [25] showed that LSTM is a practical approach for rainfall-runoff modeling. Zhang et al. [26] established a RNN, LSTM, and GRU model to forecast outflows for the Xiluodu Reservoir. Fang et al. [27] suggested that the LSTM model has better generalization in simulating soil moisture. Kratzert et al. [28] demonstrated the potential of LSTM for large-scale regional modeling in 241 basins. These research efforts show the potential of the LSTM model in illustrating the characteristics of the flood time series [29], [30], The improved model of LSTM is also widely used in prediction [31], [32]. However, LSTM research applications are still at their initial phase, with only a few studies in relatively regular disaster prediction in China.

Additionally, the quantity and quality of data on disaster causing factors affecting mountain floods are crucial, fully utilizing hydrological data can effectively improve the predictive performance of DL models at the mountain flood level forecasting [33]. When we input data to RNN, we ignored the long dependency relationship between data, the relevant data is packaged, including the previous value. In general, using the principle of similarity, the data-driven flood forecasting model has certain advantages under the DL model, with sufficient impact factor data and continuous training [34], [35]. By improving the model's sustainability and uncertainty forecasting [36], it has meaningful research and application value in small watershed mountain torrent forecasting. Big Data is importance that organizations develop the tools and means necessary for data values, and exploiting the increasingly available Big Data for their advantage [37]. In line with this, Puttinaovarat and Horkaew [38] notes that there is a need for the adoption of powerful tools such as Data Mining techniques which can aid in modelling the complex relationships that are inherent in Big Data. Big Data forecasting has the ability to improve the model's Hydrological forecasting effect, although the current predicted time scale is large [39], [40], but big Data analysis and predictive model are hand in hand on obtaining real time forecasts using the increasingly available data. However, due to the complexity of hydrological time series data prediction, there is a lack of specialized integration for multi-dimensional data, especially for hourly scale data, which affects the accuracy of DL prediction.

Specialized research gradually carries out on mountain flood forecasting, Wang et al. [41] discussed the feasibility of using data-driven and machine learning for mountain flood prediction. Gaidukova et al. [42] considered the possibilities of constructing regression equations for predicting the runoff of mountain and semi-mountain rivers. Liu and Chen [43] discussed the characteristics of mountain torrent, such as sudden, frequent, an advanced method of grid accurate early warning based on refined rainfall forecast data and measured rainfall data, can accurately predict the risk of mountain torrents 1 to 3 hours, and effectively improve the timeliness and

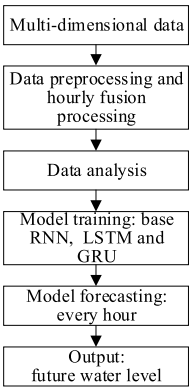


FIGURE 1. The architecture of method for mountain flood level forecasting in small watersheds.

accuracy of mountain torrents early warning. In general, the prediction of flash floods is still in the preliminary research stage, and there is great room for improvement in the accuracy and real-time performance of its quantitative prediction.

Overall, current research of DL model for forecasting is mainly limited to two points. Firstly, DL models are only used for medium to large watersheds, but small watersheds, especially mountain floods, are rarely studied due to the complexity of their water level changes. Secondly, currently only the elements of rainfall are considered, while hydrological, meteorological, geographical, and human activities are rarely mentioned due to the complexity of data processing. On the other hand, the accuracy and real-time are the main goal of the mountain flood level forecasting in small watersheds.

III. METHOD

According to the feature engineering and model training institutions of DL, the main process of our method is shown in Figure 1, which includes two parts at multi-dimensional data processing and prediction. Multi-dimensional data processing includes the acquisition of multi-dimensional disaster data, data preprocessing, hourly fusion, and data analysis; The water level prediction part includes model training, hourly forecasting by hour, and final output results. This section mainly introduces the models, including the base RNN, LSTM and GRU. The next section mainly introduces multi-dimensional data processing.

A. BASE RNN

Base RNN is feedforward neural network with temporal connections, they have states, channels and temporal connections between them. The input information of a neuron includes not only the output of the previous neuron layer, but also its own state in the previous channel. The structure of the base RNN is shown in Figure 2. Base RNN is usually also called RNN, the following are all referred to as RNN.

The hidden state (h_t) and output (y_t) equations are as follows:

$$h_t = \tanh\left(x_t W_{xh} + h_{t-1} W_{hh} + b_h\right) \tag{1}$$

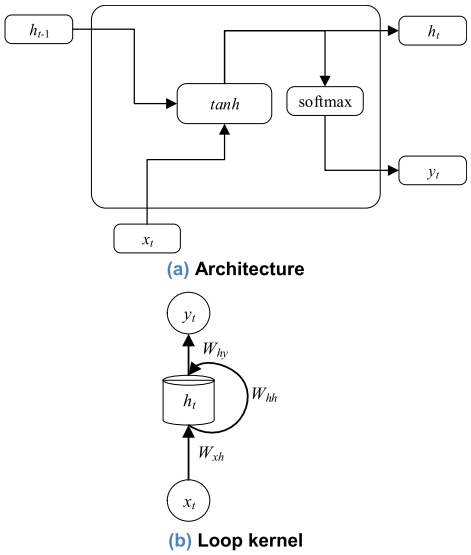


FIGURE 2. The architecture of RNN model.

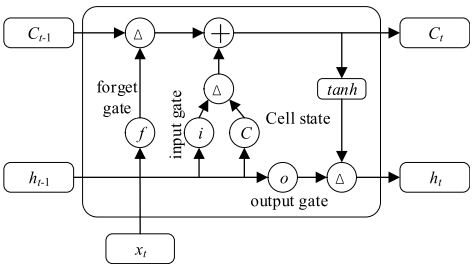


FIGURE 3. The architecture of LSTM model.

TABLE 1. LSTM model's equations.

Name	Equation
Input gate	$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$
forget gate	$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$
output gate	$O_t = \sigma(W_O x_t + U_O h_{t-1} + b_O)$
cell state of input node	$\widetilde{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$
cell state	$C_t = f_t \otimes C_{t-1} + i_t \otimes \widetilde{C}_t$
hidden state	$h_t = O_t \otimes tanh(C_t)$

$$y_t = \text{softmax}\left(h_t W_{hy} + b_y\right) \tag{2}$$

The W_{xh} is the matrix of weights between the input and hidden layers and W_{hh} is the matrix of recurrent weights between the hidden layers at adjacent time steps. The vectors b_h and b_y are biases which allow each node to learn an offset.

B. LSTM

LSTM is an adapted version of RNN to address problems of exploding or vanishing gradient and later were refined or simplified. This method can preserve information over long periods through its unique structure, cell state, and gates to control how the information flows in the different layers. The structure of the LSTM is shown in Figure 3, the gates and states' equations are in the Table 1.

TABLE 2. GRU model's equations.

Name	Equation
input	$\widetilde{x_t} = Wx_t$
forget gate	$f_t = \sigma(W_f x_t + b_f)$
reset gate	$r_t = \sigma(W_r x_t + b_r)$
cell state	$C_t = f_t \otimes C_{t-1} + (1 - f_t) \otimes \widetilde{x_t}$
hidden state	$h_t = r_t \otimes g(C_t) + (1 - r_t) \otimes x_t$

where i_t, f_t, and o_t are the input gate, the forget gate, and the output gate, respectively; W_i, W_f, and W_o represent the weights connecting input, forget, and output gates with the input, respectively; U_i, U_f, and U_o denote the weights from input, forget, and output gates to the hidden states, respectively; b_i, b_f, and b_o are input, forget, and output gate bias vectors, respectively; C_{t-1} is the cell input; C_t is the current cell state, $\tilde{C}_t$ is the intermediate input variable of C_t, h_t refers to current hidden state; σ is the logistic sigmoidal function; $\otimes$ is element-wise multiplication; and *tanh* is the hyperbolic tangent function. The intuition behind this structure design is that the cell state acts as the memory unit to remember useful information through the different operations of each gate. The input gate filter is the information that added to the cell with the assistance of current input and historical information, while the forget gate discards certain previous information, which allows the cell to clear the value it contains; at last, the output gate outputs the information selectively.

C. GRU

In order to solve the computing power of the training model, GRU can reduce computational complexity, the speed of training has been improved. Similar to LSTM, it solves the problems of long-term memory and gradients in backpropagation in RNN through gating units. Compared to LSTM, the internal network architecture of GRU is relatively simple. The GRU network contains two gates that use an update gate and a reset gate. The reset gate determines how to combine new input information with previous memory, while the update gate defines the amount of previous memory saved to the current time step. These two gate vectors determine which information can ultimately serve as the output of the gate loop unit. The special feature of these two gating mechanisms is that they can preserve information from long-term sequences and will not be cleared over time or removed due to being unrelated to prediction. The GRU model's equations are as Table 2, activation function is g, the structure of the GRU is shown in Figure 4.

IV. DATA

A. MULTI-DIMENSIONAL DATA COLLECTION

The Tiaoxi Catchment is located in the southern part of Zhejiang Province in Eastern China, which is the typical small watersheds and mountain flood prone. The main mountain range in the area is Tianmu Mountain. The terrain

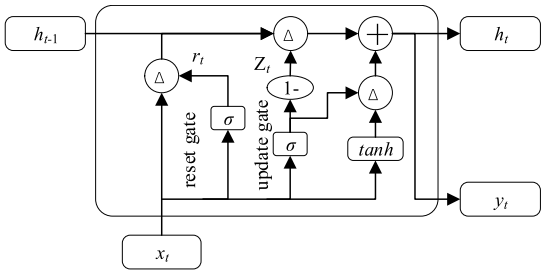


FIGURE 4. The architecture of GRU model.

gradually decreases and slopes from southwest to east and northeast. This paper choice hourly data for 1 year of typical mountain flood monitoring point, including hydrological data, meteorological data, geographical data and Human activities data in the Table 3. The hydrological data mainly come from the water conservancy management department of Zhejiang Province. Hourly meteorological data are collected from the hydrological station the China Meteorological and data Service Centre.

The inputs connect to the multi-source data of the main influencing factor data platform such as hydrology, meteorology, geography, human activities, and combine with the R&D team's water conservancy database to build a data center to achieve large-scale collection of historical, real-time and forecast data. After data preprocessing, data fusion, storage and other processes for the DL forecasting model to use, and model parameters into the model database.

In addition, large-scale instruments such as terrain scanning drone terrain collectors, multi-wave speed river terrain survey boats and other large instruments are used to collect high-density geographic data in the test area; a water conservancy data Internet of Things device is built at the test site to collect real-time hydrometeorological and geographic data to improve the real-time and accuracy of model input data. We chose data from 2411 consecutive hours to cover over a year about data from April 11th 2021 to July 20th, 2021, with the main rainfall and water levels shown in the Figure 5.

B. DATA PREPROCESSING

We use Python and other tools to perform data cleaning, conversion, and dimensionality reduction preprocessing on multi-source heterogeneous data. We use k-Nearest Neighbor (KNN) and time alignment method to extract multi-channel redundant data of the same frequency time series set, especially in the missing data section, key rainfall data is filtered through KNN using micro level data from multiple hydrological monitoring points and macro level meteorological data from monitoring areas to determine the optimal rainfall data. We fill missing data through data smoothing, perform pattern recognition on unstructured image data, transform the format and semantics of the data, and normalize the disaster-causing factor information cluster is divided into four dimensions of data including water conservancy, meteorology, geography, and crowdsourced data, and a unified semantic and data relationship ontology information model is constructed.

TABLE 3. Collection of data on the impact factors of mountain floods.

Type	Impact factors	Data characteristics	Data format	Data sources
Hydrological data	rainfall, water level, etc.	Including the main observation station in the small watershed and nearby multi-source observation stations, temporary observation stations, etc. The data changes rapidly, counting every hour to improve accuracy through redundancy.	digit, image (water gauge)	Historical data: the government department, https://sqfb.slt.zj.gov.cn/. Real time data: water conservancy data Internet of Things device is built at the test site.
Meteorological data	weather, rainfall forecast (rainfall, probability), wind direction, temperature, humidity, wind direction, wind speed, etc.	Including multiple similar forecast data sources. The data changes rapidly, counting every hour to improve accuracy through redundancy.	text, digit, image	https://data.cma.cn
Geographical data	elevation, land cover, soil erosion, etc.	Including multi-resolution, multi-channel similar data. The data has less variation and is related to the test points.	image	The government department, Internet, and terrain scanning drone terrain collectors, multi-wave speed river terrain survey boats and other large instruments are used to collect high-density geographic data in the test area.
Human activities data	buildings, transportation facilities, farming, public opinion, etc.	Has a certain degree of contingency.	Image, map, digit	The government department, Internet.

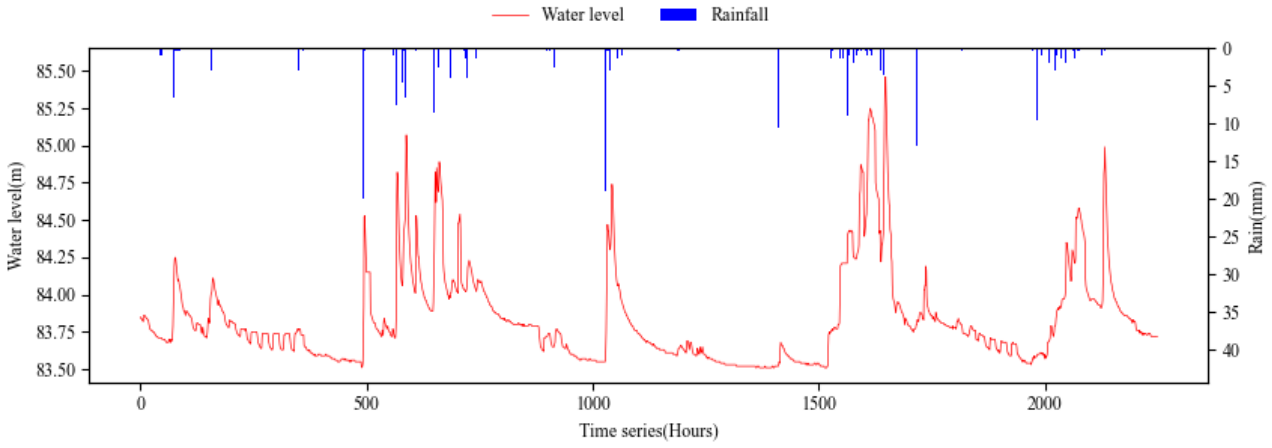


FIGURE 5. The time series of water level and rain.

According to the forecasting model, the qualitative relationship between the dedication disaster factors and the level of the disaster is constructed, and the coupling correlation between the disaster factors is analyzed through the model, the main influencing factors are extracted to reduce the input dimension and quantity of the model.

The input variables were scaled to accelerate the convergence speed and make the training process faster and were normalized using the following equation:

$$\bar{Q}_t = \frac{Q_t - Q_{min}}{Q_{max} - Q_{min}} \tag{3}$$

where Q_t and $\bar{Q}_t$ are the observed and the normalized runoff at time t, respectively.

C. DATA ANALYSIS

The water level data is the very complexity time series. In order to better compare data, we select the data sets to study, such as straight lines, sine waves, general time series information, stock price trends and noise, etc. They are compared with water level change data to determine the predictability of the data, such as Figure 6. From the graphical perspective, the water level data of the mountain flood changes rapidly and violently, and its inherent variation pattern cannot be observed.

Confusion is the standard for measuring the quality of a probability distribution or language model. Low confusion indicates that time series models can predict samples well. The calculation perplexity method is as formula (3), the result is in Table 4.

$$PP(W) = P(\omega_1\omega_2\cdots\omega_N)^{-\frac{1}{N}} = \sqrt[N]{\frac{1}{P(\omega_1\omega_2\cdots\omega_N)}} \tag{4}$$

where W is the time series data, N is the prediction length, $\omega_1, \omega_2, \ldots, \omega_N$ are the probability of prediction.

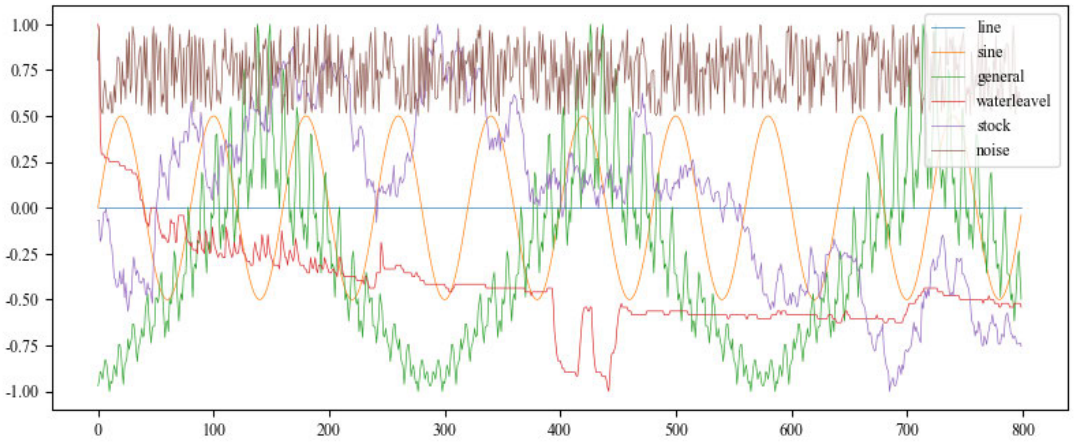


FIGURE 6. Comparison effect graph of various time series information.

TABLE 4. The perplexity of the typical time series.

No	time series	complexity	Prediction difficulty
1	Straight line	0	Very low
2	Sine function	5.69	Low
3	General time series	8.17	Medium
4	Stock price	9.03	Medium
5	Water level in this area (unusual)	11.00	High
6	Gaussian noise	12.54	High

Through quantify complexity in measured time series, the water level is approaching noise in the mountain small watersheds, which has high complexity and weak micro predictability. This is also the key reason for choosing DL models combined with multi-dimensional disaster factor data.

V. EXPERIMENT AND RESULTS

A. EXPERIMENT

Python can provide DL models, and with Python library is simple and efficient tools for predictive data analysis. Calculations were carried out with the CPU of intel(R) Xeon(R) E-2176M, double cores,2.70GHz and 32 GB RAM.

The comparative performance analysis of DL methods for long-term forecasting is presented. The model's traditional statistics measures, equations and meaning are as Table 5. Where y_i is the target output value, and $\hat{y}_i$ is the model output value. Mean Square Error (MSE) is the main metrics of the accuracy, Mean Absolute Error (MAE) reflect the robustness of the model, R-Square (R^2) and Adjusted R-Square (AR^2) reflect the performance of hydrological model.

CPU and durations are the time to use. In terms of time efficiency, record the time of the model training core step (CPU1) and the entire training stage (Duration1), as well as the time of the model prediction core step (CPU2) and the entire prediction stage (Duration2), respectively, to compare the execution efficiency of the model.

The forecasting model is constructed based on multi-dimensional water conservancy data, as shown in Figure 7. The input data include the historical and real-time data of hydrology, meteorology, geography and human activities. Through the development of a DL framework, the model can be trained and validated for prediction, so as to build a DL model scheme for each forecasting point, and finally build a model library to predict multiple monitoring points. Water level and disaster level within 24 hours, 1 to 7 days and 1 mouth as Short-Term period, Mid-Term period and Long-Term period forecasting. In this experiment, we select 112 hours for forecasting which is in the Mid-Term range, and it has strong representativeness.

Among them, X is the input; cells are including the hidden multi-layer DL units; P is the actual output; c, h are the input and forget gate parameters; L is the number of cells. The input data has been standardized and divided into training, validation, and prediction sets. The model achieves optimal parameters for each layer through repeated training, validation, and the use of Adam error training method. The real-time input data flow is predicted by the model, processed through denormalization, and finally used for predicting the water level of mountain floods.

We use the TensorFlow v2.0 framework to implement the aforementioned ensemble DL model, with the main parameters as Table 6. Additionally, it should be noted that modeling boundary conditions were kept as similar as possible to allow comparison of different models. This means that input data, number of layers, number of neurons, dropout rate, and number of training epochs were kept identical. Therefore, each model acquired a different set of parameters with the same model structure, which could be analyzed based on forecasting results. After training, the value was normalized for comparison with observed data.

B. RESULTS

We train each model 200 times and validate it simultaneously to correct the model. Finally, use it for forecasting. Figure 8 to 10 show the models' train, validation and

TABLE 5. Model's measure methods.

Name	Equation	Range	Meaning
Mean Square Error (MSE)	$MSE = \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2$	[0, +∞)	When the predicted value completely matches the true value, it is 0, which is a perfect model; the larger the error, the greater the value. It affected by extreme values.
Mean Absolute Error (MAE)	$MAE = \frac{1}{m}\sum_{i=1}^{m}\lvert y_i - \hat{y}_i\rvert$	[0, +∞)	When the predicted value completely matches the true value, it is 0, which is a perfect model; the larger the error, the greater the value. It has better robustness against outliers.
R-Square (R^2)	$R^2 = 1 - \frac{\sum_i(\hat{y}_i - y_i)^2}{\sum_i(\bar{y}_i - y_i)^2}$	[0, 1]	It is specifically used to evaluate hydrological models. 0: Poor model fitting effect; 1: The better the model fitting effect.
Adjusted R-Square (AR^2)	$AR^2 = 1 - \frac{(1-R^2)(n-1)}{n-p-1}$	[0, 1]	0: poor model fitting effect; 1: the model has no errors. The n is the number of samples and p is the number of features. Adjusted R-Square offsets the impact of sample size on R-Square, achieving true 0~1, with larger values being better.

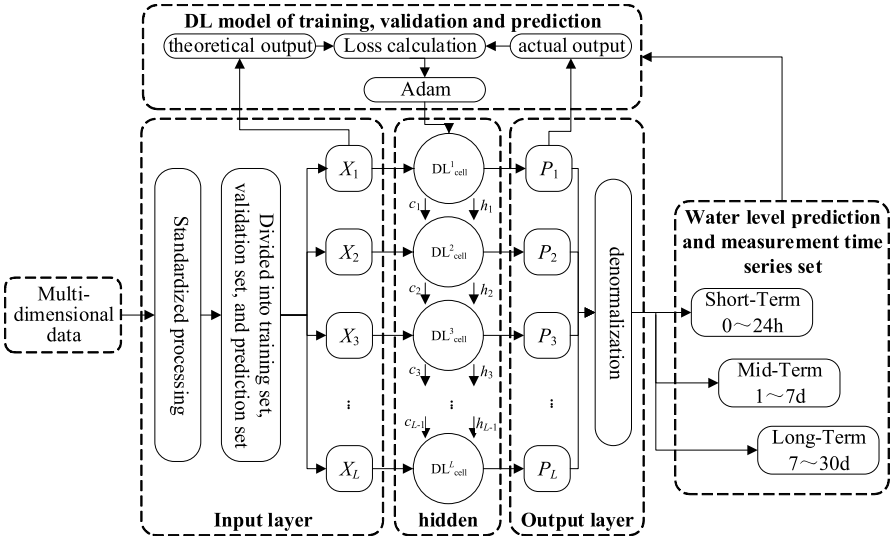


FIGURE 7. The architecture of ensemble DL system including input, hidden and output layer, the dynamic process including training, validation and prediction phase.

TABLE 6. The models' training parameters.

Parameter	Value	Meaning
units	64	Positive integer, dimensionality of the output space.
rate	0.5	The proportion of inputs that need to be discarded.
activation	relu	Activation function
kernel_regularizer	tf.keras.regularizers.l2()	Regularizer function applied to the kernel weights matrix.
adam	0.0001	Adam optimization function and value.
loss	mean_squared_error	Loss function
metrics	accuracy	List of metrics to be evaluated by the model during training and testing.
batch_size	32	Integer or None. Number of samples per batch of computation.
epochs	200	Number of epochs to train the model.

prediction. Table 7 explained the meaning and range of each data of the label in the figures. Qualitative analysis from the graphical perspective shows that time series RNN, LSTM and GRU models can be considered an efficient hydrology modeling approach. Rainfall and water level changes are basically consistent. The training and validation loss are both rapidly decrease until a stable state from the Figure (c), and the entire training performance is reliable.

At the training and validation phase, the performance is consistent. When the output of the model at the training phase has a high degree of overlap with the real water level, that is, high training accuracy, then the predicted overlap or accuracy during the validation stage is also high. It is worth noting that the LSTM has better approximation effect at the prediction phase.

We provide quantitative test data. Table 8 represents the training's loss and durations of improved models. Table 9

TABLE 7. Forecasting accuracy and durations of improved models.

Data	Meaning	Range
Water level (m)	Water level Based on the elevation of 85 Yellow Sea.	83m~86m
Time series (Hours)	Data from April 11th 2021 to July 20th, 2021.	0~2400
Water level (train_real)	80% data for training of water level.	83m~86m
Water level (train_result)	Model's output of water level corresponding to training.	83m~86m
Rain of train	Rainfall corresponding to training.	0~30mm
Water level (validation_real)	20% data for validation of water level.	83m~86m
Water level (validation_result)	Model's output of water level corresponding to validation.	83m~86m
Rain of validation	Rainfall corresponding to validation.	0~30mm
Water level (predict_real)	The real water level of the prediction time.	83m~86m
Water level (predict_result)	Model's output of predict water level.	83m~86m
Rain of predict	The rain of predict by meteorological department.	0~30mm

TABLE 8. Training accuracy and durations of improved models.

Model	MSE	MAE	R^2	AR^2	CPU1 (s)	Duration1 (s)
RNN	0.0042	0.0350	0.9590	0.9579	347.457	347.4567
LSTM	0.0041	0.0343	0.9591	0.9580	325.9978	325.9979
GRU	0.0034	0.0305	0.9663	0.9654	302.5361	302.5364

TABLE 9. Validation accuracy and durations of improved models.

Model	MSE	MAE	R^2	AR^2	CPU2 (s)	Duration2 (s)
RNN	0.0089	0.0670	0.9235	0.9021	0.1166	0.1164
LSTM	0.0057	0.0545	0.9507	0.9368	0.0957	0.0953
GRU	0.0054	0.0484	0.9541	0.9412	0.1154	0.1157

TABLE 10. Prediction accuracy and durations of improved models.

Model	MSE	MAE	R^2	AR^2	CPU2 (s)	Duration2 (s)
RNN	0.0111	0.0886	0.3549	0.1445	7.6373	7.6375
LSTM	0.0040	0.0501	0.7686	0.5894	7.5484	7.5493
GRU	0.0064	0.0613	0.6311	0.3454	7.4108	7.4110

TABLE 11. Comparison of key indicators for model forecasting.

Measure	RNN	LSTM/RNN	GRU/RNN
MSE	0.0111	35.87%	57.19%
MAE	0.0886	56.57%	69.13%
AR^2	0.3549	216.56%	177.81%
CPU2(S)	7.6373	98.84%	97.03%

represents the validation's error and durations of improved models. Table 10 represents the forecasting's error and durations of improved models. The accuracy' type include MSE, MAE, R^2 and AR^2. The durations include CPU time and the whole process time.

The three models show small differences in measured data, and Figure 11 vividly reflects the main indicators of MSE and MAE, while Table 11 provides a comparison with the predicted values. The main characteristics are as follows.

(1) In terms of MSE, LSTM has the lowest error in prediction, indicating the highest accuracy. GRU follows closely, with errors only 35.87% and 57.19% of RNN, respectively, indicating the advantage of the improved RNN in terms of accuracy.

(2) In terms of MAE, the errors of LSTM and GRU are only 56.57% and 69.13% of RNN, respectively, consistent with the characteristics of MSE. What's more, MAE reflects the stability and reliability of the entire prediction process and is not affected by outliers.

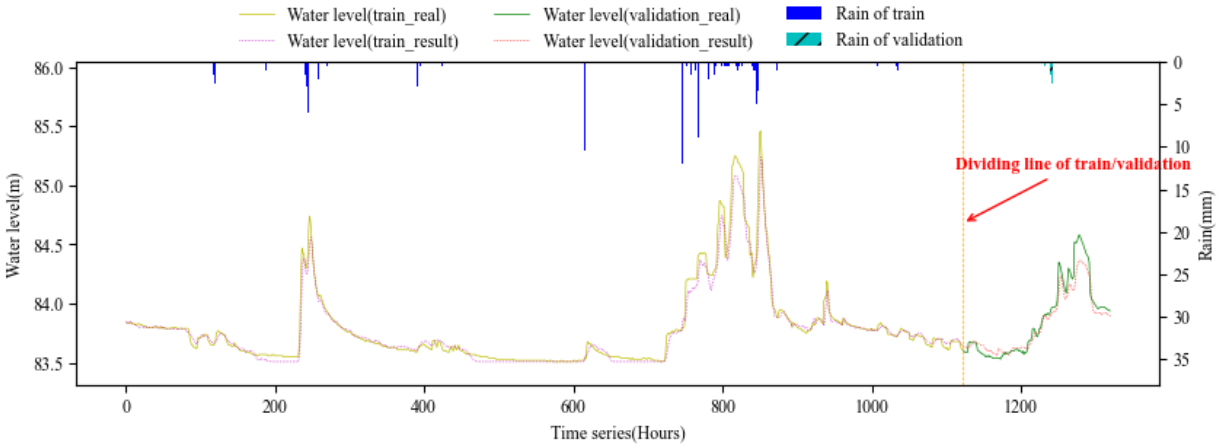
(3) In terms of R^2 and AR^2, both LSTM and GRU models have a high degree of fitting with the hydrological forecasting model, with values of 216.56% and 177.81% of RNN, respectively, LSTM performs well in hydrological prediction However, there are still limitations in the prediction phase.

(4) In terms of time consumption, the three models show little difference, but GRU has the shortest time, which is consistent with its design intention. All three models have high real-time performance, with the ability to predict a water level point in about 8 seconds, making them suitable for large-scale forecasting points in small watersheds.

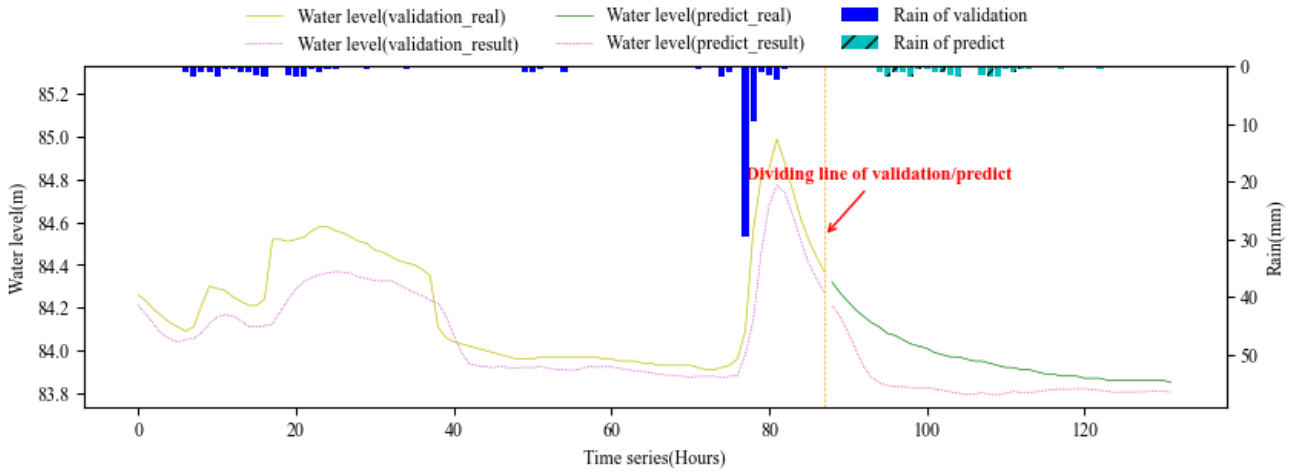
Overall, LSTM has significant advantages in terms of accuracy, stability, and real-time performance, making it an excellent method for small watershed flood forecasting. What's more, this is also dependent on the information mining and data processing of the pre-disaster factors, which sets LSTM as state-of-the-art model for mountain flood level forecasting in small watersheds.

VI. DISCUSSIONS

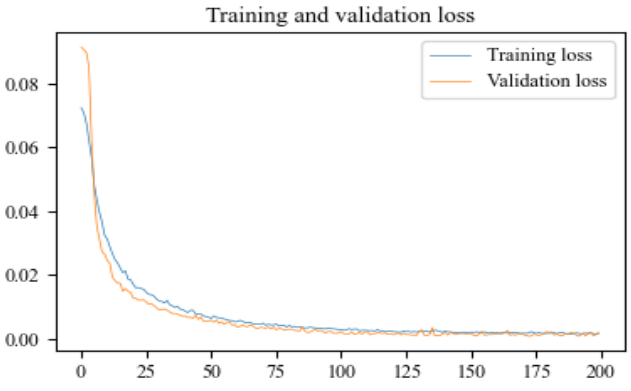
At first, we give the comparison of forecasting schemes between our method and the recent research. Then, we



(a) Training and Validation phase's water level and rain by RNN model



(b) Validation and prediction phase's water level and rain by RNN model

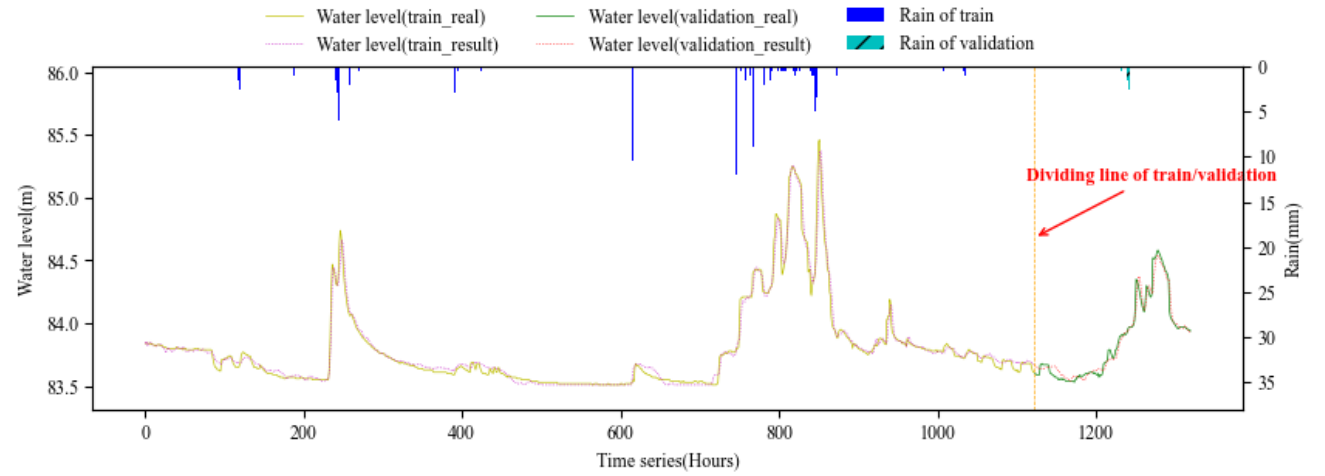


(c) Training and Validation's loss by RNN model

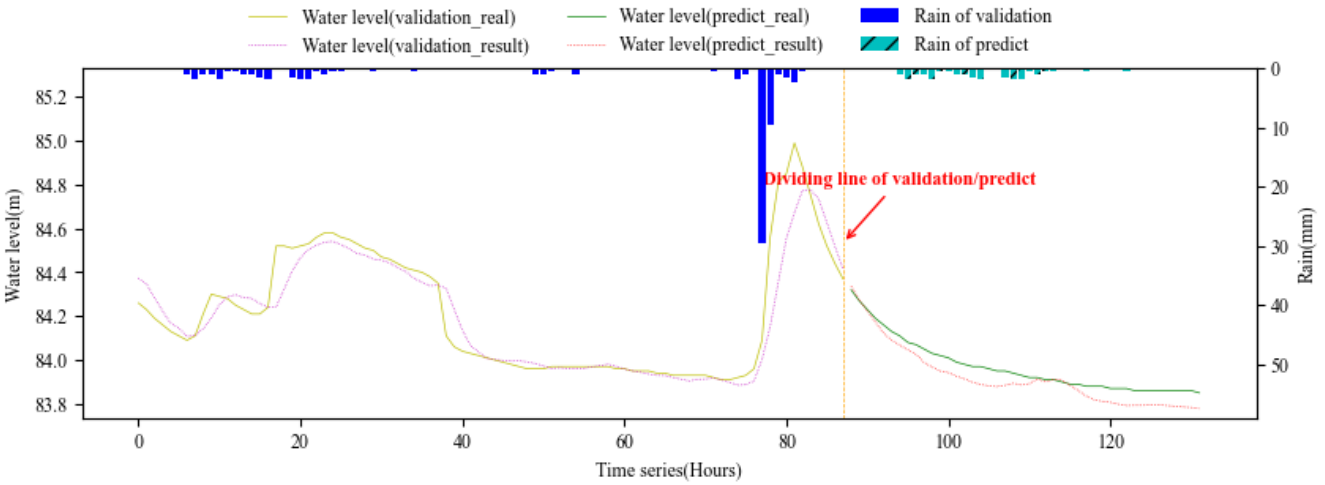
FIGURE 8. Forecasting map of mountain flood by RNN model.

discuss the data collection period and scale, model's complexity and computational efficiency, modeling approaches and model's application areas, etc**.,** there are the key factors for Mountain flood level forecasting in small watersheds based on DL. Finally, we discussed the shortcomings.

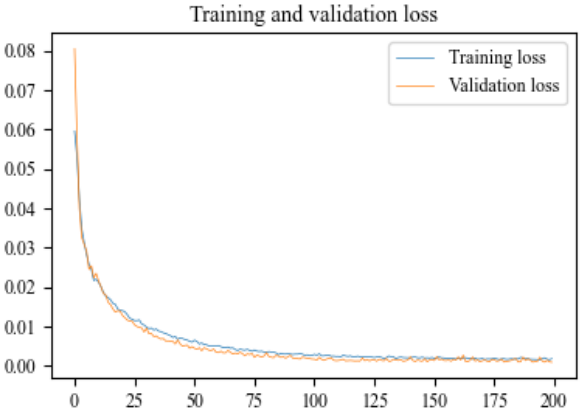
Table 12 give the comparison of forecasting schemes between our method and the recent research, including the method, area, data complexity, forecasting frequency, accuracy, real-time and computing resources. Our method the potential for mountain flood level forecasting in small watersheds. Our method of RNNs have certain advantages in the comprehensive effect of frequency, accuracy real-time and computing resources forecasting in small watersheds. Due to the early data processing and analysis, the



(a) Training and Validation phase's water level and rain by LSTM model



(b) Validation and prediction phase's water level and rain by LSTM model

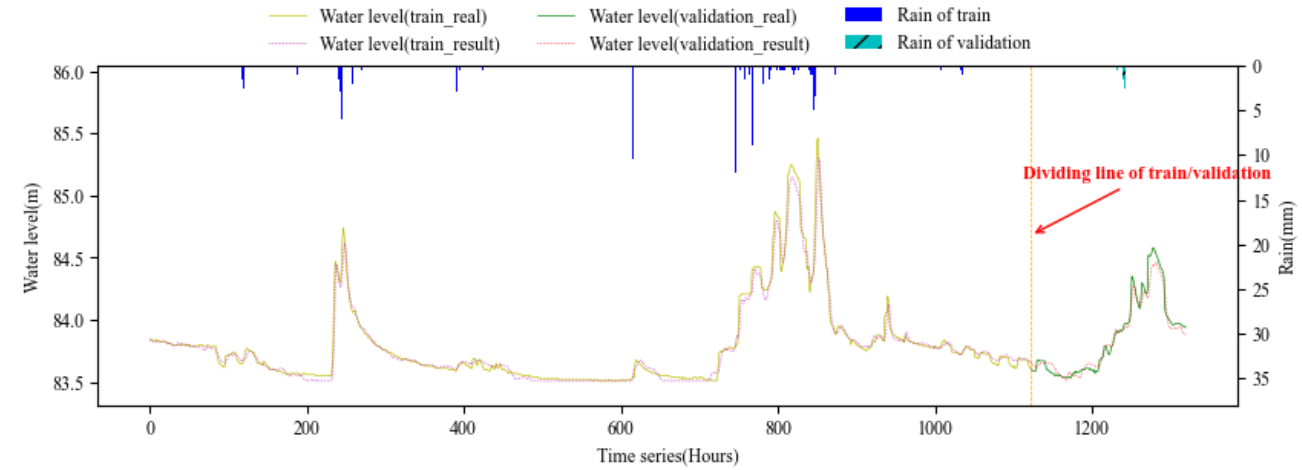


(c) Training and Validation's loss by LSTM model

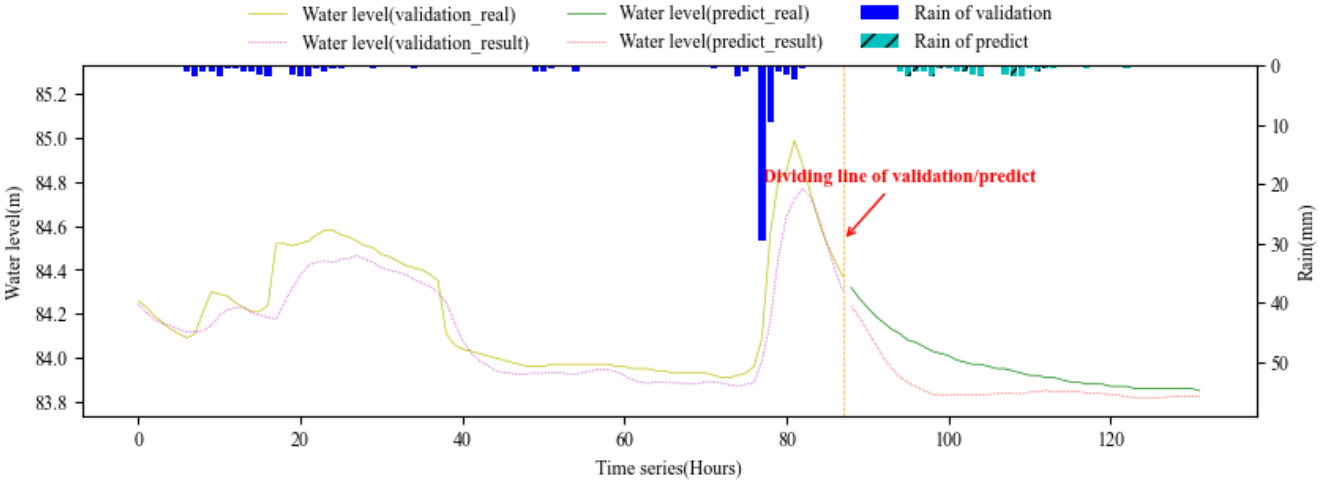
FIGURE 9. Forecasting map of mountain flood by LSTM model.

frequency and accuracy of forecasting result have been improved.

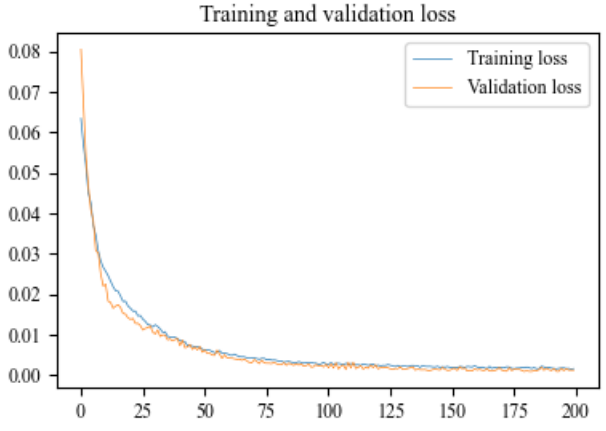
In terms of data collection multi-dimensional, period and scale, generally, the extending the data collection period can improve the model's robustness and generalizability. However, the collection of small watershed data is difficult [44], especially at the hourly level. Although the more data in the data-driven mode, the more accurate it is, however, the key



(a) Training and Validation phase's water level and rain by GRU model



(b) Validation and prediction phase's water level and rain by GRU model



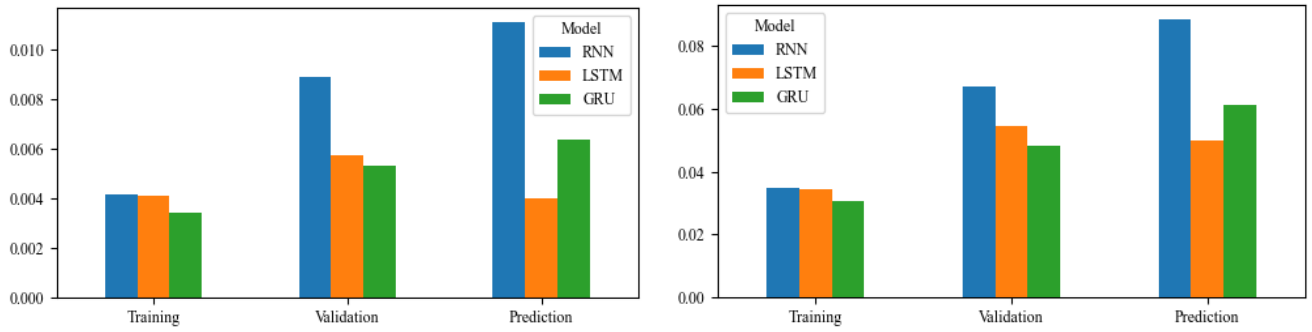
(c) Training and Validation's loss by GRU model

FIGURE 10. Forecasting map of mountain flood by GRU model.

data are only useful according to the characteristics of the traditional water conservancy model, such as the water level and rainfall in the previous cycle, the predicted rainfall, and so on. Other meteorological data seem to be small, and even affect the accuracy of water level prediction. The small amount of data used in literatures [1] and [45] can still achieve effective results under the traditional hydrological model. Especially in the field of small watersheds, there are few hydrological

TABLE 12. Comparison of forecasting schemes.

Method	Area	Data complexity	Frequency	Accuracy	Real-time	Computing resources	Ref
Our method	Mountain flood level in small watersheds	Multi-dimensional	Hourly	High	High	Low	/
Hydrological Model	Small catchments of hilly	Rainfall and water level	Hourly	Low	Medium	Low	[1][3]
Machine learning	Large watersheds	Rainfall and water level	Monthly	Medium	Low	Medium	[17] [18]
Integrated model	Universal watersheds	Rainfall and water level	Daily	High	Medium	High	[20] [21]
LSTM model	Large watersheds	Rainfall and water level	Daily	High	High	High	[29][30]



(a) MSE of training, validation and prediction of ML modes

(b) MAE of training, validation and prediction of ML modes

FIGURE 11. Evaluation chart of DL models' MSE and MAE including RNN, LSTM and GRU.

data sensing devices, and the real-time requirements of data are very high. In addition, the large amount of data affects the speed of data, especially the water level prediction of large-scale monitoring points in small watersheds. In general, the effective information content of data is the most important. Research in this area is currently lacking in data-driven DL methods.

In terms of model's complexity, in order to illustrate the complexity of changes in mountain flood water level, compared with literatures [28] and [36], the data collection period is based on year or month, and they are very regular, have low perplexity as formula (4), and relatively low prediction difficulty. So that, they are not suitable for predicting mountain flood water levels. Of course, the RNN series proposed in this article is not flawless. In order to maintain the accuracy of the model, data preprocessing is necessary, and the model needs to be continuously trained and calibrated. About modeling approaches and the experiment design, we follow the basic ideas of feature engineering, DL training, validation, and prediction, and adopt the TensorFlow v2.0 development method, which is a good DL development and production framework. Furthermore, PyTorch is an advanced experimental tool for DL model design, which better reflects the convenience of experiments.

In terms of balance between model complexity and computational efficiency, compared to the currently popular Transformer series models, it has become a popular model in Artificial General Intelligence (AGI) fields, such as ChatGPT and Sora. However, the model requires high computational resources and a large amount of data, and the computing power possessed by small or even large watersheds cannot support large models, which cannot meet the real-time requirements for predicting mountain flood water levels. In addition, the model of small watersheds does not continuously train and predict, which poses a challenge for unlimited computation. RNN series has high accuracy and real-time performance under appropriate window data, which is the original intention of this paper. Table 8 to Table 10 shows the high accuracy and real-time performance of the prediction method proposed in this paper. Compared with relevant literatures [5] and [46], the prediction of mountain floods in small watersheds is also a field that scholars rarely touch on, mainly reflected in the balance between accuracy and real-time performance, as well as the balance between model complexity and computational power.

About the model and data's coordination, through quantify complexity in measured time series, the water level is approaching noise in the mountain small watersheds, which has high complexity and weak micro predictability, the general data-driven model is difficult for forecasting, such as Auto Regressive (AR), Moving Average (MA) and Auto Regressive Integrate Moving Average (ARIMA) model which base on its own changing patterns and ignoring influencing factors. It requires sufficient exploration of input information in order to accurately predict water levels. This is also the key reason for choosing DL models combined with multi-dimensional disaster factor data.

About generalizability of the model, Due to the differences in hydrology, geography, ecology, and other environments in different small watersheds, the trained DL model cannot be

directly used for predicting mountain flood water levels in other small watersheds. Current literature has also confirmed this, and even traditional hydrological prediction models are difficult to achieve. From the perspective of water conservancy flood control and disaster reduction, accurate and real-time water level prediction has been achieved, achieving the goal. Although pure model universality is not sufficient, data preprocessing methods and modeling methods can be generalized, and we only need to retrain the model and adjust parameters.

Finally, our method also has certain shortcomings. We did not forecast the high peak and duration of rapid increase in small watersheds, which is related to the selected small watersheds. Currently, this situation is rare in the area we have chosen in the near future. With our continuous monitoring, we will further study it.

VII. CONCLUSION

This research investigated the application of the RNN models and multi-dimensional data in mountain flood forecasting. The main conclusions are as follows.

(1) By adopting RNN based models, not only can input information be obtained from the perspective of time series, but also effective information can be obtained from the perspective of horizontal spatial and spatiotemporal intersection, in order to improve the accuracy of forecasting. In response to the traditional problem of only considering rainfall as input, it has been extended to consider multiple types of data such as hydrology, meteorology, geography, human activities at hourly level., enhancing the information content of the input. The accuracy of the improved multi-dimensional sequence forecasting model has been improved, and the time factor has been enhanced during training and prediction.

(2) The experiment shows that time series RNN, LSTM and GRU models can be considered as the efficient hydrology modeling approach. The LSTM model demonstrated the best accuracy and real-time in the study areas. The GRU model requires fewer computational resources, the LSTM has high accuracy, and is suitable for predicting mountain flood water levels in large-scale small watersheds.

(3) The data and model methods designed in this article are suitable for applications in small watersheds, and data-driven RNN methods have advantages in training data and models.

In future work, we plan to challenge the forecasting of higher frequency of water level changes, denser flood periods, and longer forecasting times through experiments by composite DL model for improving Adjusted R-Square in the prediction phase. We consider conducting additional validation and modification of the models with datasets from different geographical locations to test the generalizability of the approach.

REFERENCES

[1] Z. Xin, K. Shi, C. Wu, L. Wang, and L. Ye, "Applicability of hydrological models for flash flood simulation in small catchments of hilly area in China," *Open Geosci.*, vol. 11, no. 1, pp. 1168–1181, Dec. 2019, doi: 10.1515/geo-2019-0089.

[2] W. Yuanyuan, L. Ping, S. Wenze, and Y. Xinchun, "A new framework on regional smart water," *Proc. Comput. Sci.*, vol. 107, pp. 122–128, Jan. 2017.

[3] Z. Liu, H. Zhang, and Q. Liang, "A coupled hydrological and hydrodynamic model for flood simulation," *Hydrol. Res.*, vol. 50, no. 2, pp. 589–606, Apr. 2019, doi: 10.2166/nh.2018.090.

[4] L. Wen, R. Macdonald, T. Morrison, T. Hameed, N. Saintilan, and J. Ling, "From hydrodynamic to hydrological modelling: Investigating long-term hydrological regimes of key wetlands in the Macquarie marshes, a semi-arid lowland floodplain in Australia," *J. Hydrol.*, vol. 500, pp. 45–61, Sep. 2013.

[5] S. Han and P. Coulibaly, "Bayesian flood forecasting methods: A review," *J. Hydrol.*, vol. 551, pp. 340–351, Aug. 2017.

[6] D. Li, L. Marshall, Z. Liang, A. Sharma, and Y. Zhou, "Bayesian LSTM with stochastic variational inference for estimating model uncertainty in process-based hydrological models," *Water Resour. Res.*, vol. 57, no. 9, p. 57, Sep. 2021.

[7] E. Samikwa, T. Voigt, and J. Eriksson, "Flood prediction using IoT and artificial neural networks with edge computing," in *Proc. Int. Conf. Internet Things (iThings) IEEE Green Comput. Commun. (GreenCom) IEEE Cyber, Phys. Social Comput. (CPSCom) IEEE Smart Data (SmartData) IEEE Congr. Cybermatics (Cybermatics)*, Nov. 2020, pp. 234–240.

[8] R. J. Abrahart, F. Anctil, P. Coulibaly, C. W. Dawson, N. J. Mount, L. M. See, A. Y. Shamseldin, D. P. Solomatine, E. Toth, and R. L. Wilby, "Two decades of anarchy? Emerging themes and outstanding challenges for neural network river forecasting," *Prog. Phys. Geography*, vol. 36, no. 4, pp. 480–513, 2012.

[9] K. Hsu, H. V. Gupta, and S. Sorooshian, "Artifificial neural network modeling of the rainfall–runoffff process," *Water Resour. Res.*, vol. 31, no. 10, pp. 2517–2530, 1995.

[10] Y. Seo, S. Kim, O. Kisi, and V. P. Singh, "Daily water level forecasting using wavelet decomposition and artifificial intelligence techniques," *J. Hydrol.*, vol. 520, pp. 224–243, Jan. 2015.

[11] F. J. Chang, Y. H. Tsai, P. A. Chen, A. Coynel, and G. Vachaud, "Modeling water quality in an urban river using hydrological factors—Data driven approaches," *J. Environ. Manag.*, vol. 151, pp. 87–96, Mar. 2015.

[12] Z. Alizadeh, J. Yazdi, J. H. Kim, and A. K. Al-Shamiri, "Assessment of machine learning techniques for monthly flow prediction," *Water*, vol. 10, no. 11, p. 1676, 2018.

[13] A. A. Asanjan, T. Yang, K. Hsu, S. Sorooshian, J. Lin, and Q. Peng, "Short-term precipitation forecast based on the PERSIANN system and LSTM recurrent neural networks," *JGR Atmos.*, vol. 123, no. 22, pp. 12543–12563, 2018.

[14] C. Shen, E. Laloy, A. Elshorbagy, A. Albert, J. Bales, F. Chang, S. Ganguly, K. L. Hsu, D. Kifer, Z. Fang, K. Fang, D. Li, X. Li, and W.-P. Tsai, "HESS Opinions: Incubating deep-learning-powered hydrologic science advances as a community," *Hydrol. Earth Syst. Sci.*, vol. 22, no. 11, pp. 5639–5656, 2018.

[15] Y. Lecun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, May 2015.

[16] C. A. Shen, "Transdisciplinary review of deep learning research and its relevance for water resources scientists," *Water Resour. Res.*, vol. 54, no. 11, pp. 8558–8593, 2018.

[17] F. Y. Dtissibe, A. A. Ari, H. Abboubakar, A. N. Njoya, A. Mohamadou, and O. Thiare, "A comparative study of machine learning and deep learning methods for flood forecasting in the Far-North region, Cameroon," *Sci. Afr.*, vol. 23, Mar. 2024, Art. no. e02053.

[18] S. Vidya, S. Gayathri, M. Dilsha, and S. Krithika, "Rainfall based flood prediction in Kerala using machine learning," *Int. J. Intell. Syst. Appl. in Eng.*, vol. 12, no. 16s, pp. 141–144, 2024.

[19] A. Shahabi and N. Tahvildari, "A deep-learning model for rapid spatiotemporal prediction of coastal water levels," *Coastal Eng.*, vol. 190, Jun. 2024, Art. no. 104504.

[20] J. Liu, J. Xiong, Y. Chen, H. Sun, X. Zhao, F. Tu, and Y. Gu, "An integrated model chain for future flood risk prediction under land-use changes," *J. Environ. Manag.*, vol. 342, Sep. 2023, Art. no. 118125, doi: 10.1016/j.jenvman.2023.118125.

[21] J. Li, V. Dao, K. Hsu, B. Analui, J. D. Knofczynski, and S. Sorooshian, "Improving cascade reservoir inflow forecasting and extracting insights by decomposing the physical process using a hybrid model," *J. Hydrol.*, vol. 630, Feb. 2024, Art. no. 130623.

[22] M. S. Jahangir, J. You, and J. Quilty, "A quantile-based encoder–decoder framework for multi-step ahead runoff forecasting," *J. Hydrol.*, vol. 619, Apr. 2023, Art. no. 129269.
[23] Y. D. Jhong, C. S. Chen, B. C. Jhong, C. H. Tsai, and S. Y. Yang, "Optimization of LSTM parameters for flash flood forecasting using genetic algorithm," *Water Resour. Manag.*, vol. 38, pp. 1141–1164, Jan. 2024.
[24] J. Zhang, Y. Zhu, X. Zhang, M. Ye, and J. Yang, "Developing a long short-term memory (LSTM) based model for predicting water table depth in agricultural areas," *J. Hydrol.*, vol. 561, pp. 918–929, 2018.
[25] C. Hu, Q. Wu, H. Li, S. Jian, N. Li, and Z. Lou, "Deep learning with a long short-term memory networks approach for rainfall-runoffff simulation," *Water*, vol. 10, no. 11, p. 154, 2018.
[26] D. Zhang, Q. Peng, J. Lin, D. Wang, X. Liu, and J. Zhuang, "Simulating reservoir operation using a recurrent neural network algorithm," *Water*, vol. 11, no. 4, p. 865, 2019.
[27] K. Fang, C. Shen, D. Kifer, and X. Yang, "Prolongation of SMAP to spatiotemporally seamless coverage of continental U.S. using a deep learning neural network," *Geophys. Res. Lett.*, vol. 44, no. 21, pp. 11–30, 2017.
[28] F. Kratzert, D. Klotz, C. Brenner, K. Schulz, and M. Herrnegger, "Rainfall-runoffff modelling using long short-term memory (LSTM) networks," *Hydrol. Earth Syst. Sci.*, vol. 22, no. 11, pp. 6005–6022, 2018.
[29] H. Chu, Z. Wang, and C. Nie, "Monthly streamflow prediction of the source region of the yellow river based on long short-term memory considering different lagged months," *Water*, vol. 16, no. 4, p. 593, 2024.
[30] J. B. Renteria-Mena, D. Plaza, and E. Giraldo, "Multivariate hydrologycal modelling based on long short term memory networks for water level forecasting," *Preprints*, pp. 1–23, Apr. 2024, doi: 10.20944/preprints202404.0260.v1.
[31] A. Parsaie, R. Ghasemlounia, A. Gharehbaghi, A. Haghiabi, A. A. Chadee, and M. R. G. Nou, "Novel hybrid intelligence predictive model based on successive variational mode decomposition algorithm for monthly runoff series," *J. Hydrol.*, vol. 634, May 2024, Art. no. 131041.
[32] Y. Li, X. Lei, B. Wang, K. Tan, and Z. Jiang, "Enhancing water level prediction of the Yangtze river based on RGL-stacking model," *SSRN J.*, pp. 1–20, 2024.
[33] E. Snieder, R. Shakir, and U. T. Khan, "A comprehensive comparison of four input variable selection methods for artificial neural network flow forecasting models," *J. Hydrol.*, vol. 583, Apr. 2020, Art. no. 124299.
[34] L. Chen, N. Sun, C. Zhou, J. Zhou, Y. Zhou, J. Zhang, and Q. Zhou, "Flood forecasting based on an improved extreme learning machine model combined with the backtracking search optimization algorithm," *Water*, vol. 10, no. 10, p. 1362, Sep. 2018.
[35] W. Li, A. Kiaghadi, and C. Dawson, "Exploring the best sequence LSTM modeling architecture for flood prediction," *Neural Comput. Appl.*, vol. 33, pp. 5571–5580, Jun. 2021.
[36] D. Li, L. Marshall, Z. Liang, A. Sharma, and Y. Zhou, "Characterizing distributed hydrological model residual errors using a probabilistic long short-term memory network," *J. Hydrol.*, vol. 603, Dec. 2021, Art. no. 126888.
[37] S. Puttinaovarat and P. Horkaew, "Application programming interface for flood forecasting from geospatial big data and crowdsourcing data," *Int. J. Interact. Mob. Technol.*, vol. 13, no. 2019, pp. 137–156, 2019.
[38] S. Puttinaovarat and P. Horkaew, "Flood forecasting system based on integrated big and crowdsource data by using machine learning techniques," *IEEE Access*, vol. 8, pp. 5885–5905, 2020.
[39] M. R. Sheikh and P. Coulibaly, "Review of recent developments in hydrologic forecast merging techniques," *Water*, vol. 16, no. 2, p. 301, 2024.
[40] S. Swagatika, J. C. Paul, B. B. Sahoo, S. K. Gupta, and P. K. Singh, "Improving the forecasting accuracy of monthly runoff time series of the brahmani river in India using a hybrid deep learning model," *J. Water Climate Change*, vol. 15, no. 1, pp. 139–156, Jan. 2024.
[41] X. Wang, P. Gourbesville, and C. Liu, "Flash floods: Forecasting, monitoring and mitigation strategies," *Water*, vol. 15, no. 9, p. 1700, 2023.
[42] E. V. Gaidukova, V. G. Margaryan, and I. O. Vinokurov, "Possibilities of application of logistic regression in hydrological forecasts (on the example of the mountain river Samur)," in *Proc. SHS Web Conf.*, vol. 164, 2023, pp. 1–4.
[43] W. Liu and S. Chen, "Study on mountain torrent warning method based on refined rainfall forecast," in *Proc. 2nd Int. Conf. Digit. Soc. Intell. Syst. (DSInS)*, Apr. 2023, p. 125991J.
[44] S. Wang and O. Xu, "Semantic information modeling and implementation method for water conservancy equipment," *IEEE Access*, vol. 11, pp. 133879–133890, 2023, doi: 10.1109/ACCESS.2023.3336817.
[45] B. Szeląg, D. Majerek, A. L. Eusebi, A. Kiczko, F. de Paola, A. McGarity, G. Wałek, and F. Fatone, "Tool for fast assessment of stormwater flood volumes for urban catchment: A machine learning approach," *J. Environ. Manage.*, vol. 355, Mar. 2024, Art. no. 120214.
[46] I. Braud et al., "The challenges of flash flood forecasting," in *Mobility in the Face of Extreme Hydrometeorological Events 1*. Elsevier, 2018, pp. 63–88.



SONGSONG WANG received the Ph.D. degree in mechanical engineering from Zhejiang Sci-Tech University, Hangzhou, Zhejiang, China, in 2019. Currently, he is a full-time Teacher with the College of Information Engineering, Zhejiang University of Water Resources and Electric Power, and a Postdoctoral Researcher with the Ocean University of China. He is involved in teaching and research work of smart water and AI, this research funded by the Ministry of Education Humanities and Social Sciences of China "Research on the Probability Deep Learning Early Warning Decision Scheme for Mountain Floods Considering Social Factors" and Joint Funds of Zhejiang Provincial Natural Science Foundation of China.



OUGUAN XU received the B.S. degree in chemical engineering and process and the M.S. degree in chemical engineering from Zhejiang University of Technology, Hangzhou, Zhejiang, China, in 2000 and 2003, respectively, and the Ph.D. degree in control science and engineering from Zhejiang University, Hangzhou, in 2007. From 2007 to 2021, he was a Lecturer, a Vice Professor, and a Professor with Zhijiang College, Zhejiang University of Technology. He is currently a Professor with Zhejiang University of Water Resources and Electric Power, Hangzhou. His research interests include AI, digital twins, smart water conservancy, complex industrial process control and optimization, and soft sensor application.